

```
import pandas as pd
```

```
df = pd.read_csv('/content/vaccination_all_tweets.csv', encoding='latin-1')
```

```
print(df.shape)
```

```
df.head()
```

```
(193343, 16)
```

```
/tmp/ipykernel_1824/3841727601.py:3: DtypeWarning: Columns (8,15) have mixed types. Specify dtype option
```

```
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```

	id	user_name	user_location	user_description	user_created	user_follower
0	1340539111971516416	Rachel Roh	La Crescenta-Montrose, CA	Aggregator of Asian American news; scanning di...	2009-04-08 17:52:46	405.
1	1338158543359250433	Albert Fong	San Francisco, CA	Marketing dude, tech geek, heavy metal & '80s ...	2009-09-21 15:27:30	834.
2	1337858199140118533	eliðð±ððð'ððð°ðððððððð	Your Bed	heil, hydra ððððâð°	2020-06-25 23:30:28	10.
3	1337855739918835717	Charles Adler	Vancouver, BC - Canada	Hosting "CharlesAdlerTonight" Global News Radi...	2008-09-10 11:28:53	49165.
4	1337854064604966912	Citizen News Channel	NaN	Citizen News Channel bringing you an alternati...	2020-04-23 17:58:42	152.

```
!pip install vaderSentiment
```

```
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer  
analyzer = SentimentIntensityAnalyzer()
```

```
Requirement already satisfied: vaderSentiment in /usr/local/lib/python3.12/dist-packages (3.3.2)  
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from vaderSentiment) (2.31.0)  
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests) (3.3.2)  
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests) (3.6)  
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests) (2.2.3)  
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests) (2024.7.4)
```

```
import re
```

```
def clean_tweet(text):  
    text = re.sub(r'http\S+', '', str(text))  
    text = re.sub(r'@\w+', '', text)  
    text = re.sub(r'#', '', text)  
    text = text.strip()  
    return text
```

```
df['clean_text'] = df['text'].apply(clean_tweet)  
df[['text', 'clean_text']].head()
```

	text	clean_text
0	Same folks said daikon paste could treat a cyt...	Same folks said daikon paste could treat a cyt...
1	While the world has been on the wrong side of ...	While the world has been on the wrong side of ...
2	#coronavirus #SputnikV #AstraZeneca #PfizerBio...	coronavirus SputnikV AstraZeneca PfizerBioNTec...
3	Facts are immutable, Senator, even when you're...	Facts are immutable, Senator, even when you're...
4	Explain to me again why we need a vaccine @Bor...	Explain to me again why we need a vaccine wh...

```
def get_sentiment(text):  
    score = analyzer.polarity_scores(text)['compound']
```

```

    if score >= 0.05:
        return 'Positive'
    elif score <= -0.05:
        return 'Negative'
    else:
        return 'Neutral'

df['sentiment'] = df['clean_text'].apply(get_sentiment)
df['sentiment'].value_counts()

```

	count
Neutral	98386
Positive	64461
Negative	30496

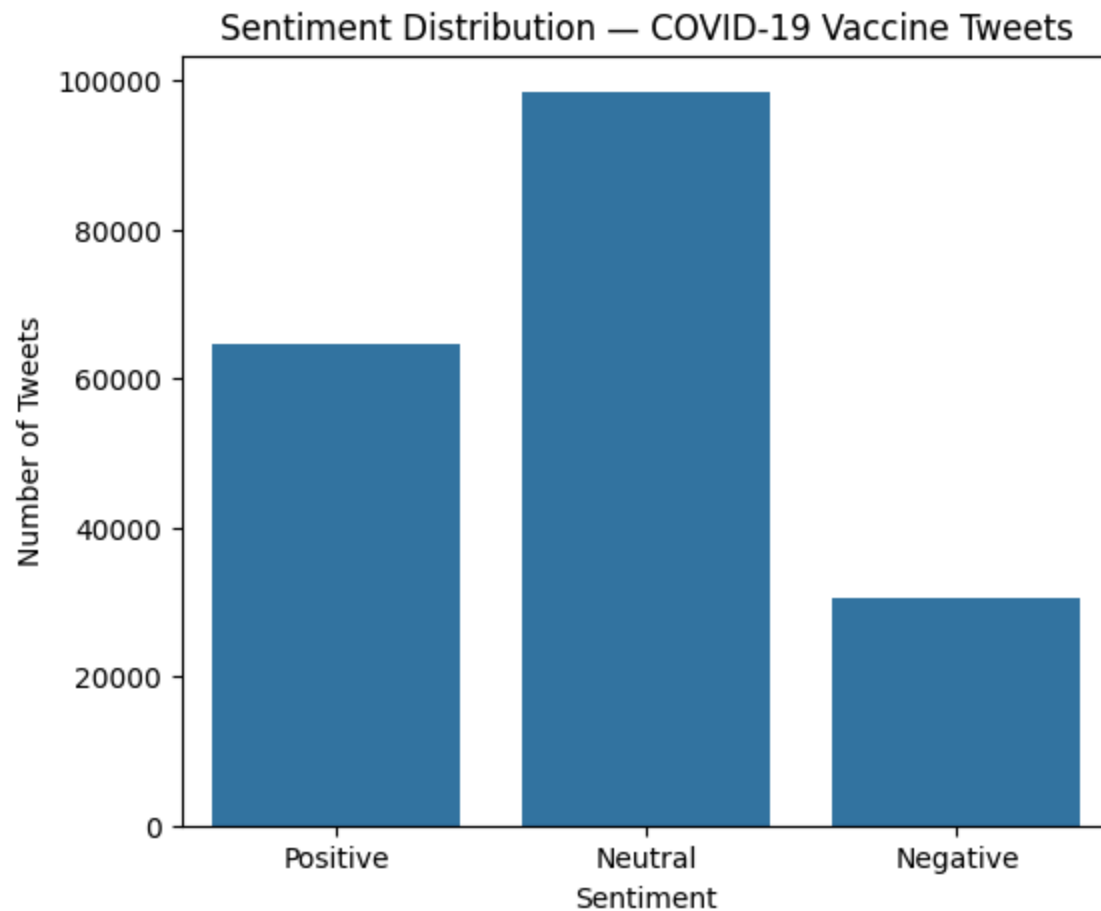
dtype: int64

```

import matplotlib.pyplot as plt
import seaborn as sns

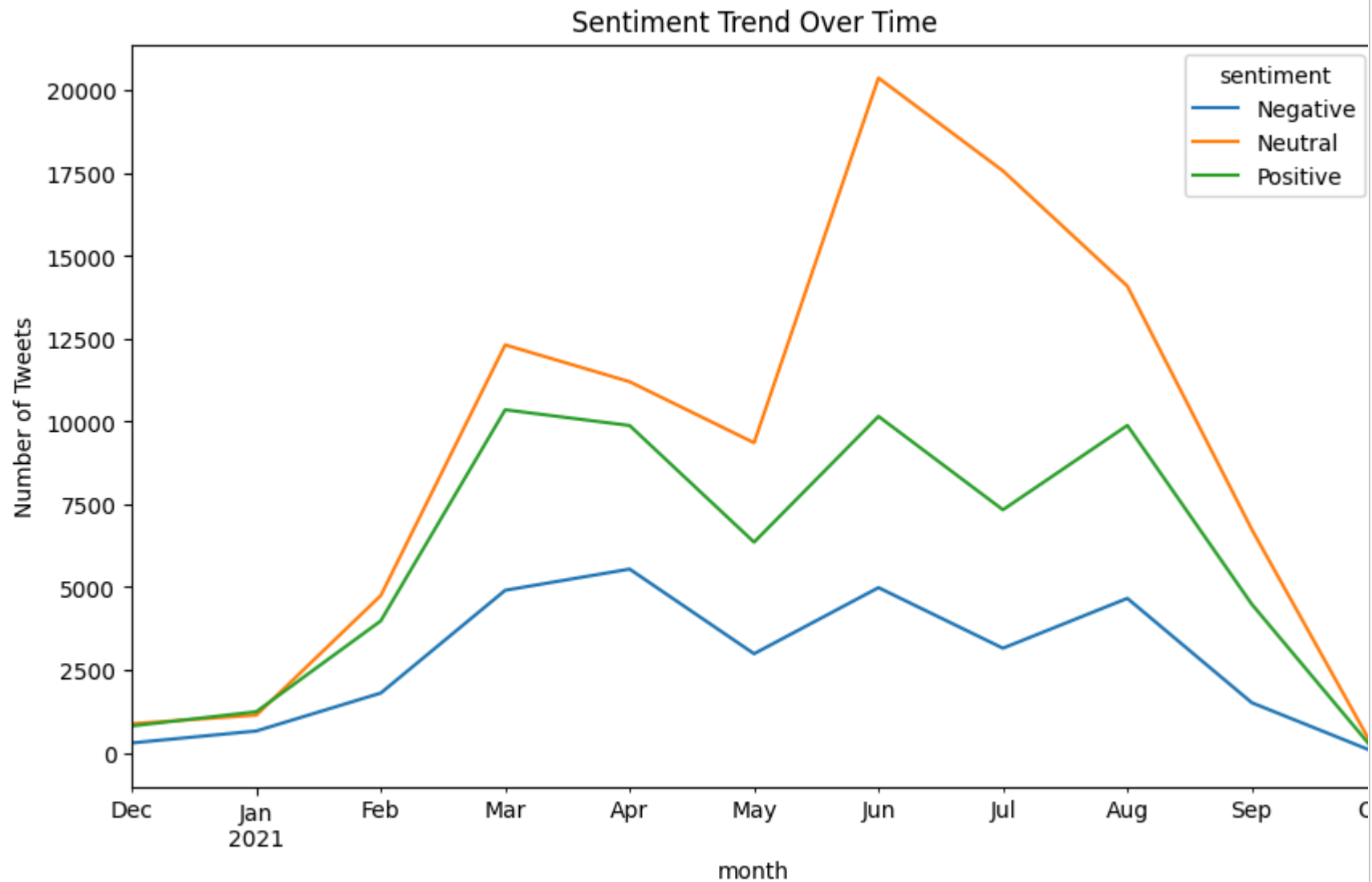
plt.figure(figsize=(6,5))
sns.countplot(data=df, x='sentiment', order=['Positive','Neutral','Negative'])
plt.title('Sentiment Distribution – COVID-19 Vaccine Tweets')
plt.xlabel('Sentiment')
plt.ylabel('Number of Tweets')
plt.show()

```



```
df['date'] = pd.to_datetime(df['date'])
df['month'] = df['date'].dt.to_period('M')

trend = df.groupby(['month', 'sentiment']).size().unstack().fillna(0)
trend.plot(kind='line', figsize=(10,6))
plt.title('Sentiment Trend Over Time')
plt.ylabel('Number of Tweets')
plt.show()
```



```
print("Sample Positive tweets:")
print(df[df['sentiment']=='Positive']['clean_text'].sample(3).to_string(index=False))

print("\nSample Negative tweets:")
print(df[df['sentiment']=='Negative']['clean_text'].sample(3).to_string(index=False))
```

Sample Positive tweets:
18-44 RURAL Bengaluru CovidVaccine Availability...
You can have your Pfizer or Moderna mRNA myocar...
No disrespect to other manufacturers but I genu...

Sample Negative tweets:
560047, \nAGE 18+\n42-D1, 39-D2 slots,\nCOVAXIN...
Yup - Got my 2nd Moderna shot and it kicked my ...
I am beyond grateful that I got my second dose ...

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Key Findings

****1. Overall sentiment distribution:**** 33% positive, 51% neutral, 16% negative — public sentiment toward COVID-19 vaccines was predominantly neutral, with positive sentiment more than double negative sentiment.

****2. Sample positive tweets**** expressed themes like vaccine availability and appointment updates, while ****negative tweets**** referenced side effects, distrust of specific manufacturers, and access/slot issues.

****3. Sentiment trend over time:**** all three sentiment categories peaked around May-June 2021, likely reflecting the period of highest public discussion during vaccine rollout, then declined toward October as the topic became less prominent on Twitter.

****4. Limitations:**** VADER is a lexicon-based tool, so it can miss sarcasm, context, and mixed sentiment. It's also tuned

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